

CSCI 3120: Operating Systems

Summer 2003

Instructor:

Kirstie Hawkey

Email:

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Office hours (outside Room 311):

Mon: 2:30-3:30, Fri: 10:30-11:30

Lectures:

MWF 1:35-2:25 PM, CS Auditorium

Course Web Page: www.cs.dal.ca/~hawkey/3120/

Prerequisites

- **CSCI 2121**
 - you need to know about hardware and CPU instructions
- **CSCI 2110**
 - because you need data structures
- **CSCI 2131 or CSCI 2132**
 - because you need to know C
- **At least C- required in prerequisite courses**
- **Prerequisites will be enforced**
 - Prerequisite waivers will only be accepted from the Undergraduate Advisor, Dr. Mike McAllister

Textbooks

- **Required:**

- *Operating Systems: Operating System Concepts (Sixth Edition)*, Silberschatz, Galvin & Gagne, Wiley, 2002.
 - *available in Dal Bookstore*

- **Other Resources:**

- *The C Programming Language*, Kernighan and Ritchie, Prentice Hall.
- *Advanced Programming in the UNIX Environment*, Stevens, Addison Wesley

Evaluation

- Midterm test 1 (in-class): 15%
- Midterm test 2 (in-class): 15%
- Final Exam (scheduled by registrar): 40%
- Assignments (top 5 of 6): 20%
- Paper: 10%
- Students are responsible for all material covered in class, and must be aware of all announcements made in class
- **Assignment/paper marks will only be included in the final mark if the combined score on the midterms and final exam is more than 35 out of 70.**
 - (I.E.: Need 50% on exam/test portion to pass course...)
- **There are no supplemental exams in this course.**

Assignments

- 6 assignments – top 5 marks counted.
- Some will involve programming (on borg), and **programming assignments must be written in C.**
- Assignments are due at 1:35 PM on the due date (i.e. just **before** class time)
- Late assignments will **not** be marked (and will therefore count as zero). So if you have not finished an assignment, hand it in anyway!
- All work handed in **must be your own work.** Please read and understand the university policy on plagiarism (**p.23-24 in Calendar**). **Offences will be reported to and dealt with by the Senate Discipline Committee.**

About Copying....

- Assignments are designed to help you learn some concept
- If you “figure it out” and do it yourself, you will learn the concept
- Assignment marks will be based upon the demonstrated understanding of the material.
- It is unacceptable to cut and paste other people’s solutions, even if attributed to their original source.

Re: Helping each other

- You may discuss homework in a general way with other students...
- but it is **not** OK to consult any one else's written work.
- Any similarity in form or notation between submissions with different authors will be regarded as evidence of copying -- so **protect your work**.
- If someone copies your solution to an assignment, we won't know which was the original so **both the original author and the person doing the copying** will end up being penalized.

Assignment Processing

- All programming assignments will be submitted electronically.
- The Teaching Assistant(s) will mark the assignments for content.
- In parallel, I will compare the assignments with each other to identify and examine the ones that appear to be the same or substantially similar.
- Assignments which appear to have been copied and/or doctored will be sent to the Senate Discipline Committee.

Research Paper – OS comparison

- Group paper, 2-4 students per group
- 1 student per OS
- Each group contains at least one flavour of Windows and Linux
- 7.5% based on individual contribution, 2.5% based on group analysis
- Paper will be incrementally assigned. Feedback will be given for sections submitted with the assignment.

Important Dates:

- **Last day to add:**
 - May 16
- **Room change (room 3157 Dentistry):**
 - May 26 - June 6
- **Last day to drop (w/o a “W”):**
 - June 2
- **Midterm Tests (tentative dates)**
 - Wednesday, June 4(regular class time)
 - Friday, July 4 (regular class time)
- **Last day to drop (with a “W”):**
 - June 30
- **Final Exam (3 hour, scheduled by registrar):**
 - August 5-9

Planned Topics:

- Introduction (Chapters 1-3, read on your own)
- Processes
- Scheduling
- Threads
- Concurrency, synchronization, deadlock
- Memory Management
- Virtual Memory
- File Systems, Disk Scheduling (time permitting)

Up next...

- Overview of operating systems
- This week read Chapters 1-3
- For Monday, read Chapter 4